conference. Aside from the DeWitts, John Wheeler was there, as was his student and Everett's friend Charles Misner. Feynman was there too, but he registered under a pseudonym, "Mr. Smith," to protest the sorry state of research in the field. At the conference, Feynman and physicist Hermann Bondi presented closely related ironclad arguments that finally convinced the physics community that gravitational waves must be real if general relativity was correct, clearing up a major embarrassment for the still-nascent research field. (This also kicked off a sixty-year search for gravitational waves, ultimately culminating in the first successful detection of gravitational waves at LIGO, a pair of four-kilometer-long laser-powered gravitational wave observatories, in 2015—an accomplishment that earned a Nobel Prize for Thorne and two of his colleagues in 2017.) And Wheeler, meanwhile, pushed his agenda of "radical conservatism," taking the predictions of established theories seriously even in wildly strange, untested, and remote domains—for example, the small, hot, dense period in the universe's history, immediately after the still-controversial Big Bang, when both general relativity and quantum physics would be necessary for understanding the behavior of the universe.

In the 1960s, the new field accelerated significantly. New mathematical techniques led to the realization that collapsed stars—or as John Wheeler dubbed them in 1968, "black holes"—must be real. And in 1964, two physicists at Bell Labs, Arno Penzias and Robert Wilson, stumbled upon a radio hiss coming from all directions in the sky, and realized that they had discovered the CMB radiation, the oldest light in the universe and an echo of the Big Bang. Within fifteen years, the steady-state theory had lost all credibility, the Big Bang model was accepted as basically correct, and Penzias and Wilson had shared a Nobel Prize. There was still a great deal of disagreement about basic questions like the rate at which the universe was expanding, but relativistic cosmology was finally off and running, with a shared model of how the universe as a whole behaves.

But the rise of cosmology made the inadequacy of the Copenhagen interpretation more acute. How can you draw a line between the observer and the observed system, as Bohr had required, when the system in question is the entire universe? "Quantum gravity is certainly going to be important in the early moments of the universe, then you are driven to the notion of a wave function for the universe, and how do you interpret a thing like that